SoGraB

A Visual Method for Soft Grasping Benchmarking and Evaluation

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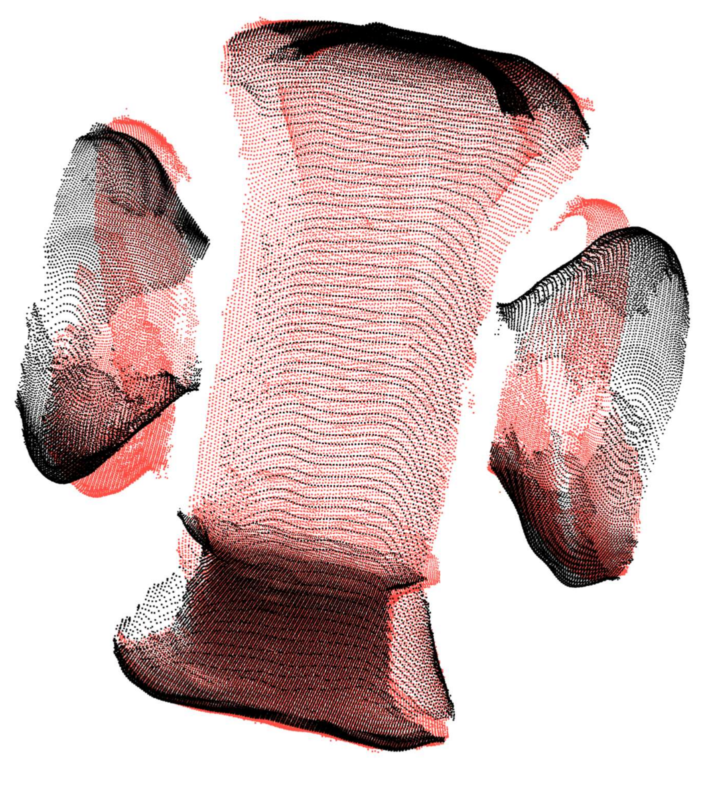
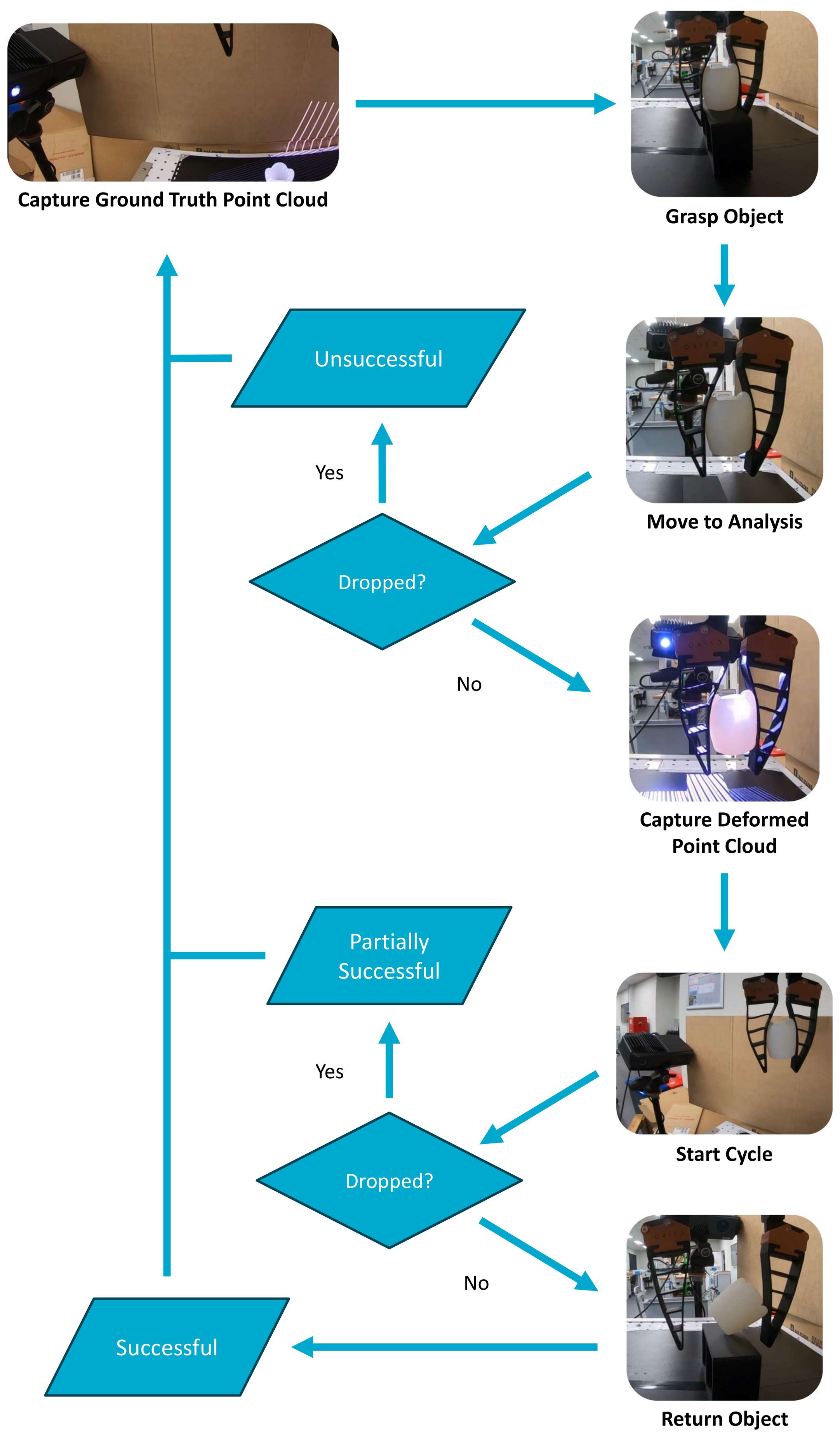


Recent years have seen soft robotic grippers gain increasing attention due to their ability to robustly grasp soft and fragile objects. However, a commonly available standardised evaluation protocol has not yet been developed to assess the performance of varying soft robotic gripper designs. This work introduces a novel protocol, the Soft Grasping Benchmarking and Evaluation (SoGraB) method, to evaluate grasping quality, which quantifies object deformation by using the Density-Aware Chamfer Distance (DCD) between point clouds of soft objects before and after grasping. We validated our protocol in extensive experiments, which involved ranking three Fin-Ray gripper designs with a subset of the EGAD object dataset. The protocol appropriately ranked grippers based on object deformation information, validating the method's ability to select soft grippers for complex grasping tasks and benchmark them for comparison against future designs.

Methodology



Methodology for determining the SoGraB score of a gripper design. This chart represents a single trial. A minimum of five trials is recommended for each gripper-object pair to calculate statistically significant mean scores and standard deviations.

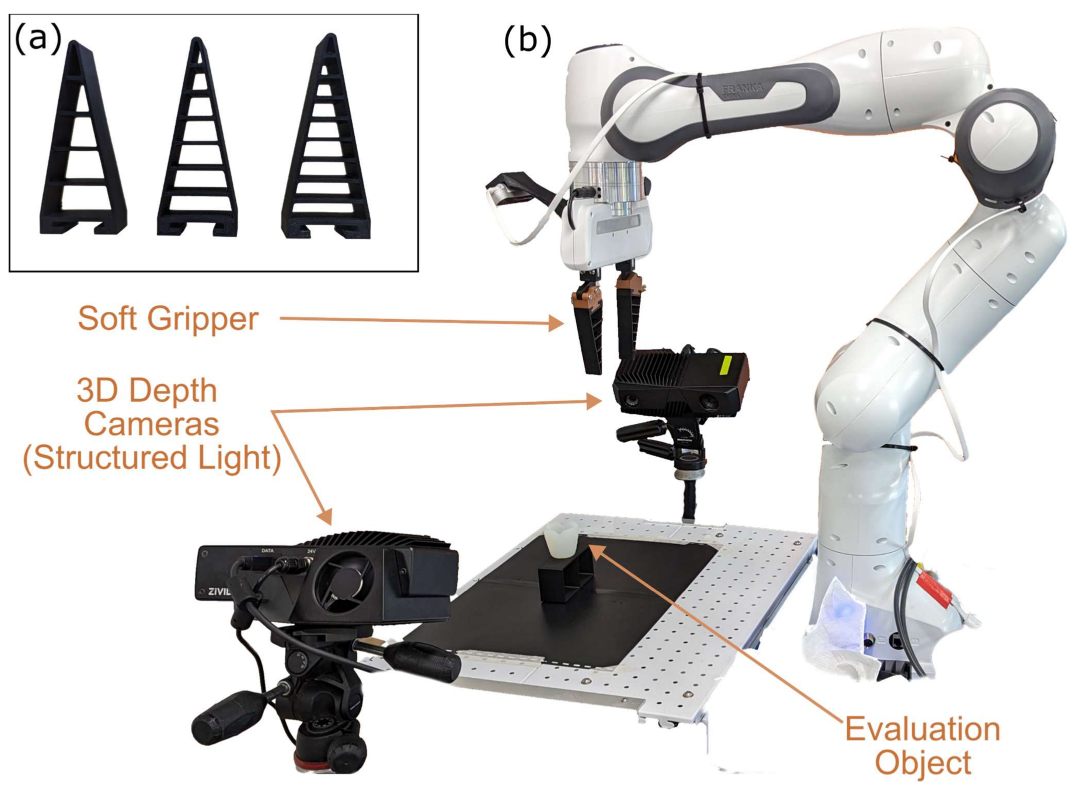
$$\text{score} = \begin{cases} 0 & \text{Unsuccessful grasp} \\ \frac{(1-d_{\text{DCD}})t_{\text{dropped}}}{2t_{\max}} \in [0, 0.5] & \text{Partially successful grasp} \\ 1 - \frac{d_{DCD}}{2} \in [0.5, 1] & \text{Successful grasps} \end{cases}$$

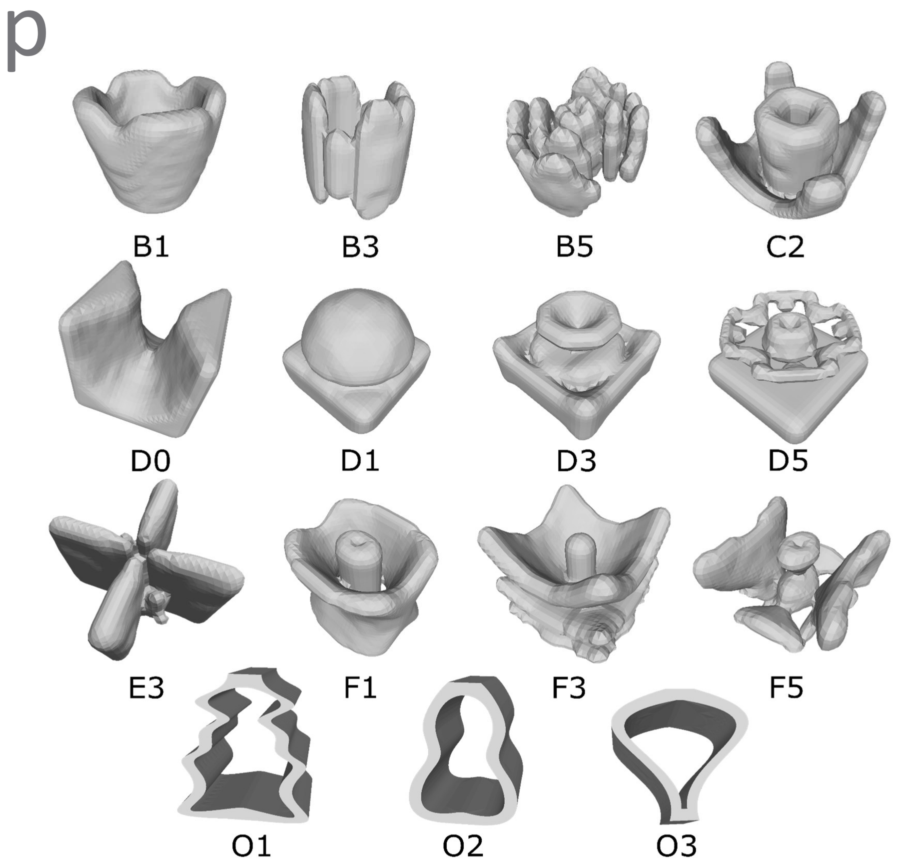
The SoGraB score assesses grasp success, holding time and object deformation – higher scores represent better performance.

$$d_{\text{DCD}}(S_1, S_2) = \frac{1}{2}\left(\frac{1}{|S_1|}\sum_{x \in S_1}\left(1 - \frac{1}{n_{\hat{y}}}e^{-\alpha\|x-\hat{y}\|_2}\right) + \frac{1}{|S_2|}\sum_{y \in S_2}\left(1 - \frac{1}{n_{\hat{x}}}e^{-\alpha\|y-\hat{x}\|_2}\right)\right)$$

Object deformation is determined by the Density-Aware Chamfer Distance (DCD) [1] between point clouds taken before and during a grasp for each object.

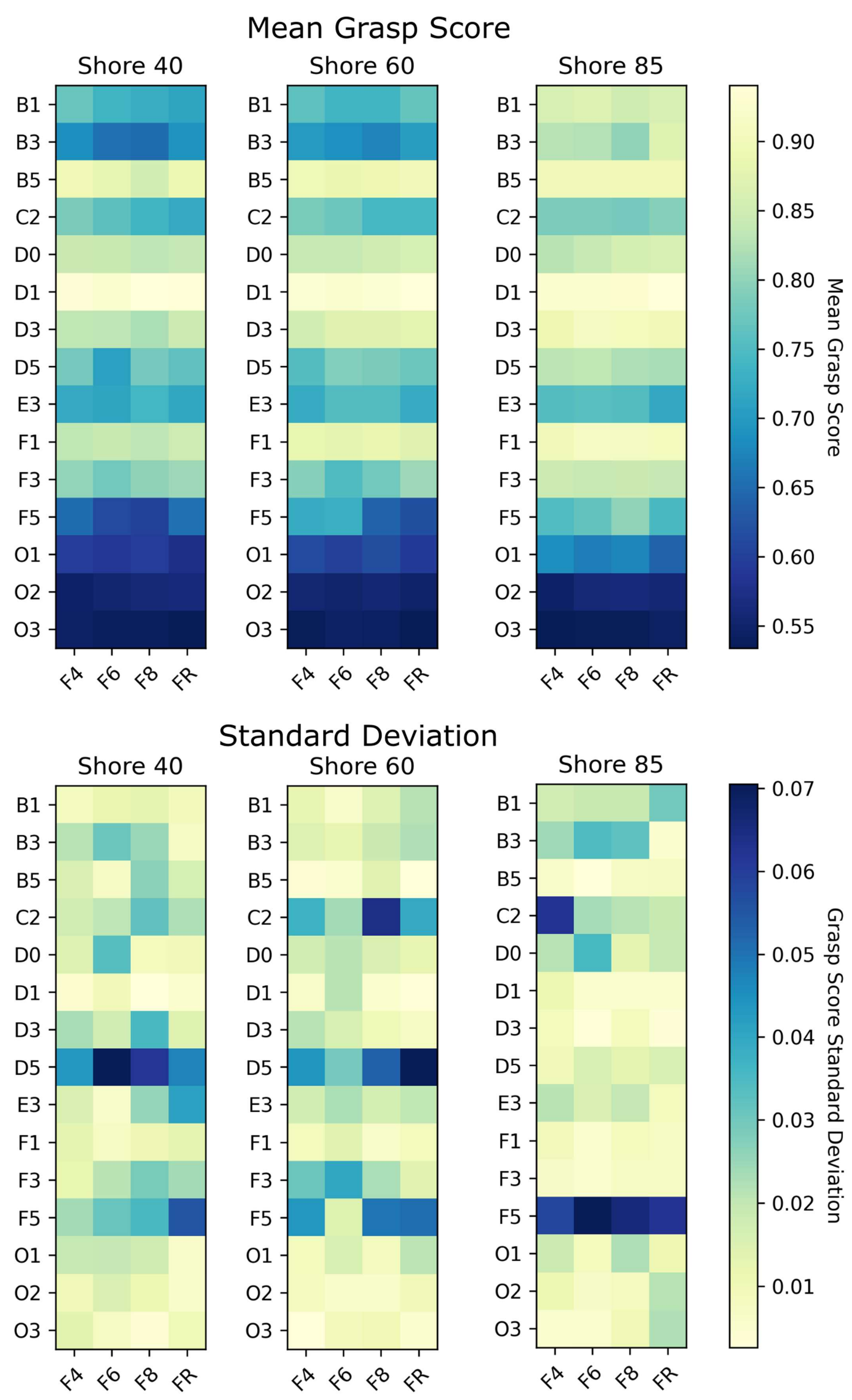
Setup

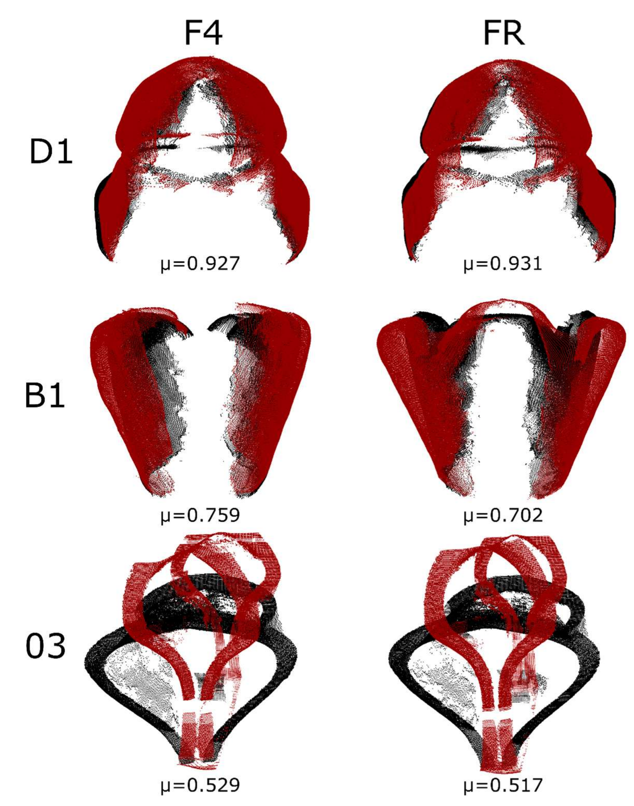




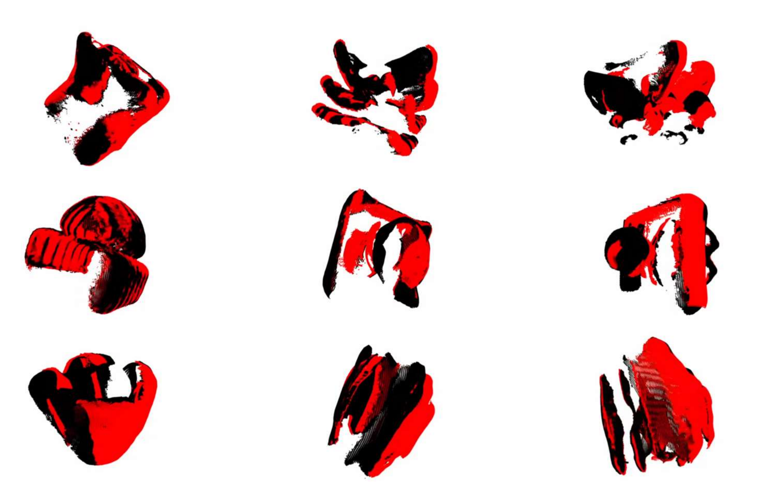
Our experimental grasp evaluation platform for assessing the performance of different soft gripper designs (a) and the three Fin-Ray designs that were evaluated (b). We used a subset of 12 objects from the EGAD dataset, in addition to three custom objects (O1-O3), printed in three different shore hardnesses. These were tested against four gripper designs, including three Fin-ray designs with various number of ribs and one rigid design to act as a baseline.

Experimental Results





We can rank different gripper designs with commonly accessible equipment! Our results show a more rigid grippers cause more object deformation then their more compliant counterparts. For new grippers, any design can be evaluated so long as the object remains at least partially visible.



Future Work

Through the ongoing use of SoGraB we aim to improve the quality of soft gripper designs and identify the most valuable use cases for soft gripping. Future users can contribute to the dataset by running the SoGraB protocol. You can: expanded the range of objects to evaluate new shapes and hardnesses, or benchmark new soft grippers against the existing object dataset.

FOR FURTHER INFORMATION
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REFERENCES

[1] T. Wu, L. Pan, J. Zhang, T. WANG, Z. Liu, and D. Lin, "Chamfer Distance as a Comprehensive Metric for Point Cloud Completion," in Advances in Neural Information Processing Systems, vol. 34. Curran Associates, Inc., 2021, pp. 29 088–29 100.